

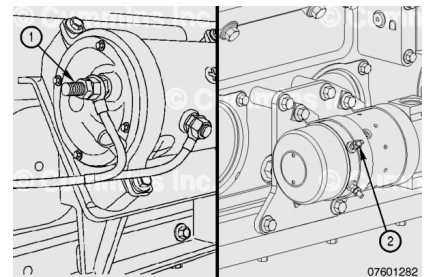
Trainings ▾

General Information

There are 2 pre-lube pump designs.

One type is referred to as the "High Volume" PRELUB™ pump. It is distinguished by the ground connection located in the end of the pump body (1).

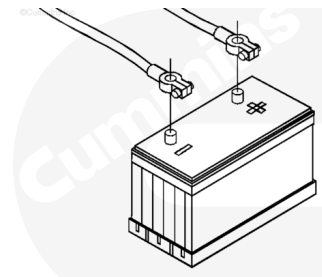
The other type is referred to as the "Compact" PRELUB™ pump. It is identified by having both of the electrical connections located on the side of the pump body (2).



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ WARNING ⚠

On some applications, switches labeled battery disconnect will not disconnect the battery power from the engine starters. Make certain that the battery power has been removed from the engine starters by checking with a

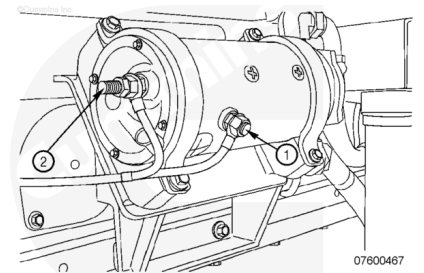
multimeter. There must be zero voltage at the starter before beginning this procedure. If there is voltage at the starter, then disconnect the battery terminals directly, removing the negative (-) battery cable first. This must be done to reduce the possibility of personal injury.

Disconnect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>)

Remove

The following explains the removal of the "High Volume" PRELUB™ pump.

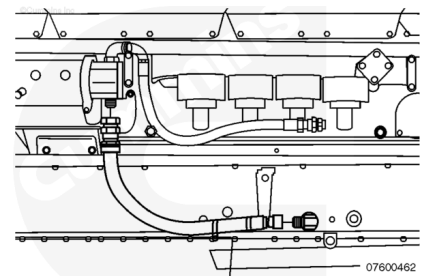
Remove the post timer relay wire (1) and the ground wire (2) from the PRELUB™ pump.



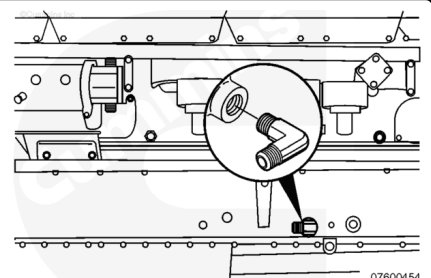
Disconnect the suction hose from the PRELUB™ pump.

Disconnect the other end of the suction hose from the elbow located on the oil pan.

Remove the oil pan capscrew, p-clip, and suction hose.

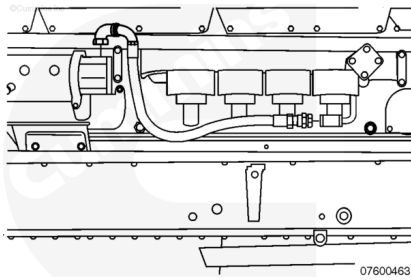


Remove the suction hose elbow.



Disconnect the supply hose from the PRELUB™ pump.

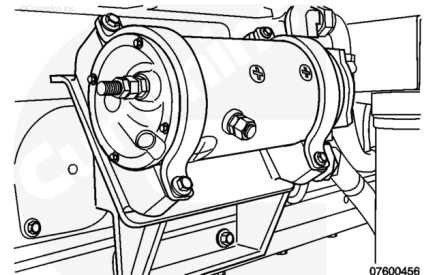
Disconnect the supply hose from the male adapter in the filter head.



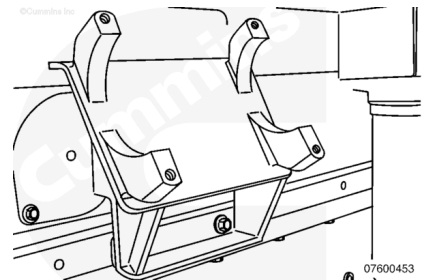
Remove the upper U-shaped pump bracket capscrews.

Loosen the lower U-shaped pump bracket capscrews until the PRELUB™ pump can be removed.

Remove the capscrews and U-shaped pump brackets.

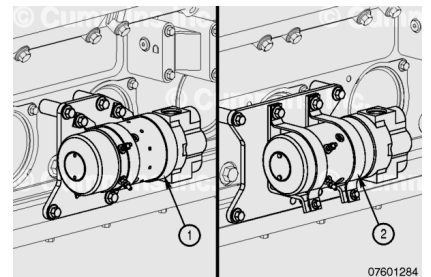


Remove the PRELUB™ pump bracket mounting capscrews and remove the bracket.



The following steps explain the removal of the "Compact" PRELUB™ pump.

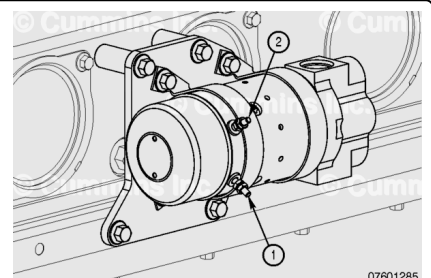
The compact PRELUB™ pump can be fitted with alternative brackets and clamps, depending on the specific application. The location of the pump on the engine may also vary, but the same procedures can be applied. Two of the most common installations are shown in the illustration.



1. Single piece 'loop' clamp
2. 2 piece 'U shaped' clamp

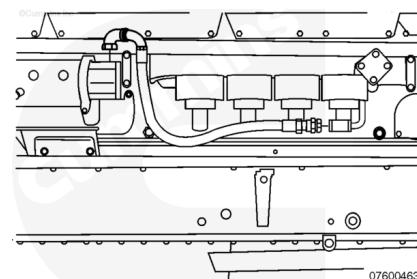
Remove the positive wire (1) and the ground wire (2) from the PRELUB™ pump.

Note : The orientation of the pump may vary. The end of the positive connection stud is marked with red paint when new. If this is no longer visible, it is essential to identify which connection is the positive connection, and mark it accordingly to allow for correct assembly.

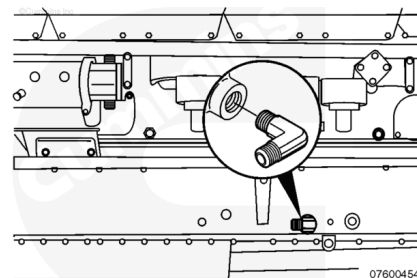


Disconnect the suction hose from the PRELUB™ pump.

Remove any hose clips fitted and disconnect the other end of the suction hose from the fitting located in the oil pan.

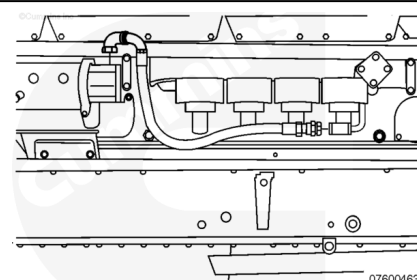


Remove the suction hose elbow.



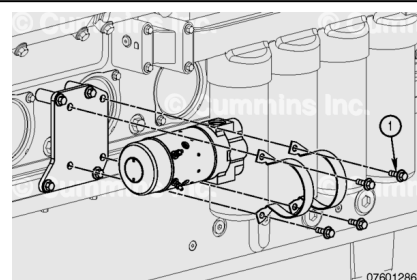
Disconnect the supply hose from the PRELUB™ pump.

Remove any hose clips fitted and disconnect the other end of the suction hose from the fitting located in the oil pan.



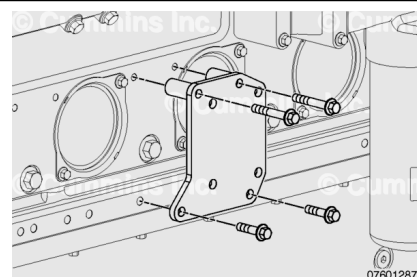
Loosen the clamping capscrews (1) and remove the pump from the engine.

Remove the clamping straps from the mounting bracket.



Remove the capscrews that retain the mounting bracket to the engine block.

Remove the mounting bracket.



Install

The following explains the installation of the "High Volume" PRELUB™ pump.

Install the bracket and tighten the capscrews.

Torque Value:

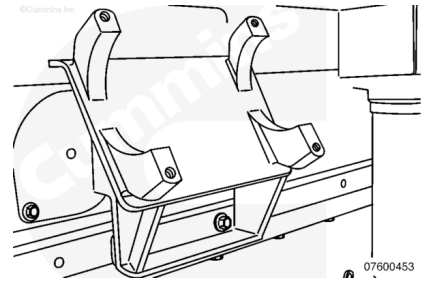
M12

1.80 n•m [59 ft-lb]

Torque Value:

M10

1.45 n•m [33 ft-lb]



Apply Loctite™ 641 to the capscrew threads.

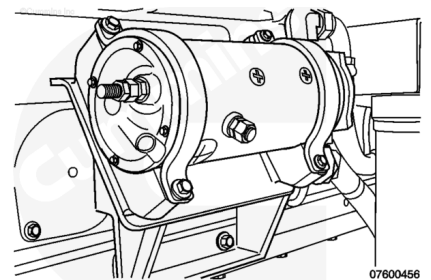
Place the PRELUB™ pump into the machined blocks of the bracket.

Place the two U-shaped brackets over the PRELUB™ pump.

Install the capscrews and tighten in an alternating sequence.

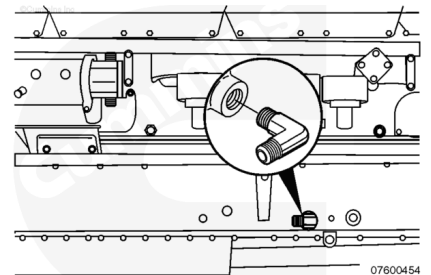
Torque Value: 30 n•m [22 ft-lb]

Make sure the gap is even under each side of the clamp.



Install the elbow into the oil pan and tighten. The final position of the elbow after it has been tightened will be facing toward the rear of the engine.

Torque Value: 74 n•m [55 ft-lb]



Connect the supply hose to the PRELUB™ pump.

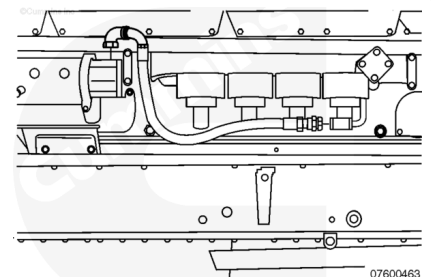
Connect the other end of the hose to the filter head male adapter.

Tighten the fittings.

Torque Value: 163 n•m [120 ft-lb]

Attach the hose to the hose bracket and tighten the capscrew.

Torque Value: 45 n•m [33 ft-lb]



Connect the suction hose to the PRELUB™ pump.

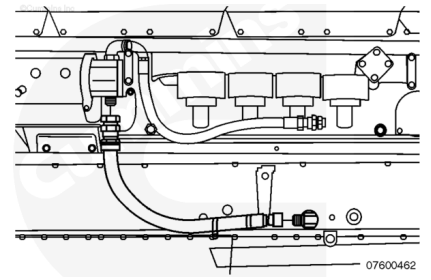
Connect the other end of the suction hose to the elbow in the oil pan.

Tighten the fittings.

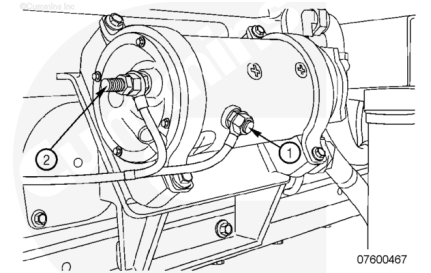
Torque Value: 190 n•m [140 ft-lb]

Attach a p-clip to the hose and secure it to an oil pan adapter capscrew.

Torque Value: 45 n•m [33 ft-lb]



Install the post timer relay wire (1) and the ground wire (2) to the PRELUB™ pump.



The following steps explain the installation of the "Compact" PRELUB™ Pump Installation

Compact" PRELUB™ Pump Installation

- Install the bracket and tighten the capscrews.

Torque Value:

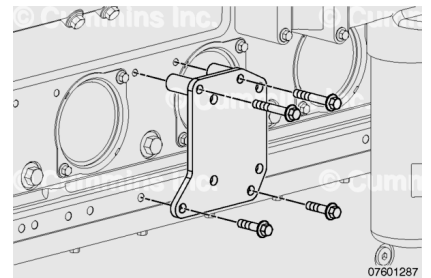
M12

1. 80 n•m [59 ft-lb]

Torque Value:

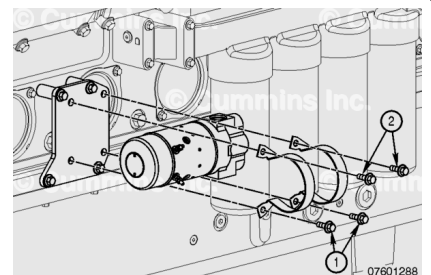
M10

1. 45 n•m [33 ft-lb]



If the single piece 'loop' clamps are installed then loosely fit the lower capscrews (1) retaining the clamps to the mounting bracket. Install the pump into the clamps. Install the upper capscrews (2). Tighten the 4 retaining capscrews in an alternating sequence.

Torque Value: 45 n•m [33 ft-lb]

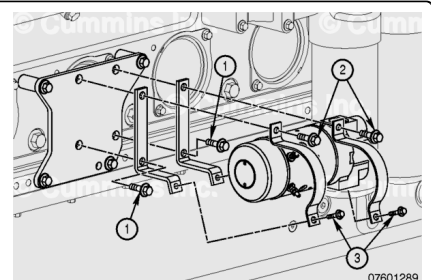


If the 2 piece 'U shaped' clamp is installed fit the clamps to the mounting bracket.

Tighten **only** the lower capscrews (1).

Torque Value: 45 n•m [33 ft-lb]

Install the pump into the clamps.



Loosely install the nuts and bolts that join the ends of the clamps (3).

Tighten the upper capscrews (2).

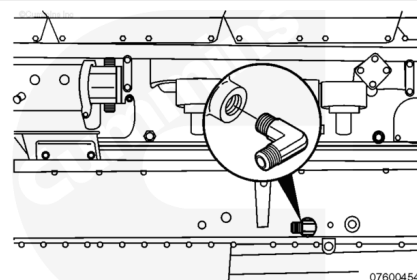
Torque Value: 45 n•m [33 ft-lb]

Tighten the clamping nut and bolt (3).

Torque Value: 30 n•m [22 ft-lb]

Install the elbow into the oil pan and tighten. The final position of the elbow, after it has been tightened, **must** be correct for the suction tube routing.

Torque Value: 74 n•m [55 ft-lb]

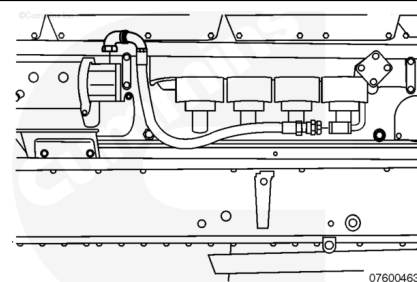


Connect the supply hose to the PRELUB™ pump.

Connect the other end of the hose to the filter head male adapter.

Tighten the fittings.

Torque Value: 163 n•m [120 ft-lb]



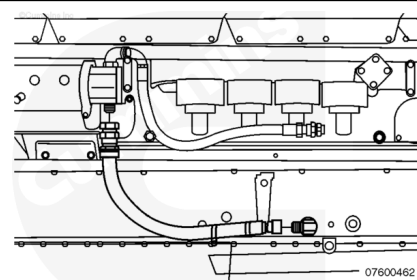
Connect the suction hose to the PRELUB™ pump.

Connect the other end of the suction hose to the elbow in the oil pan.

Tighten the fittings.

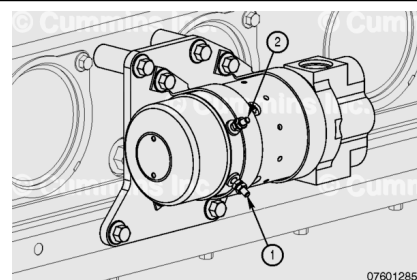
Torque Value: 190 n•m [140 ft-lb]

Replace any hose clips that were removed.



Install the positive wire (1) and the ground wire (2) to the PRELUB™ pump.

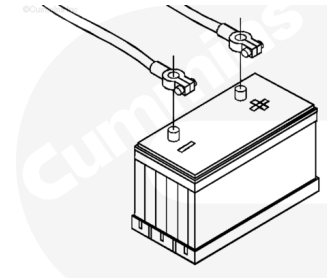
Note : Make sure of correct polarity for the connections. The pump may be orientated in the opposite direction as compared to the illustration.



Finishing Steps

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Connect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>)